

Set 16: Customer Demographics Analysis

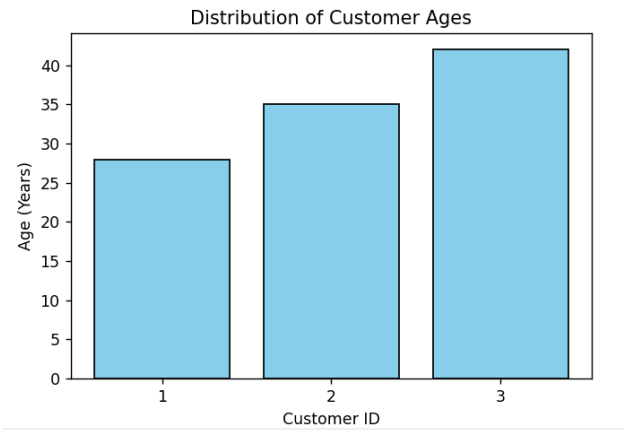
Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

- 1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
- 2. Generate a pie chart to represent the distribution of customers by gender.
- 3. Build a table to show the demographic information for each customer. Label the table elements.

Code:

```
# -----  
  
# Set 16: Customer Demographics Analysis  
# -----  
  
import pandas as pd  
import matplotlib.pyplot as plt  
  
# -----# Create  
Dataset  
# -----  
  
data = {  
    "Customer ID": [1, 2, 3],  
    "Age": [28, 35, 42],  
    "Gender": ["Female", "Male", "Female"],  
    "Income ($)": [50000, 60000, 75000]  
}  
  
df = pd.DataFrame(data)  
  
# Display Dataset
```



```

print("Customer Demographics Dataset")
print(df)

# -----
# 1. Bar Chart - Customer Ages
# -----

plt.figure(figsize=(6,4))

plt.bar(df["Customer ID"].astype(str),
        df["Age"],
        color="skyblue",
        edgecolor="black")

plt.title("Distribution of Customer Ages")
plt.xlabel("Customer ID")
plt.ylabel("Age (Years)")

plt.savefig("Customer_Ages_BarChart.png")
plt.show()

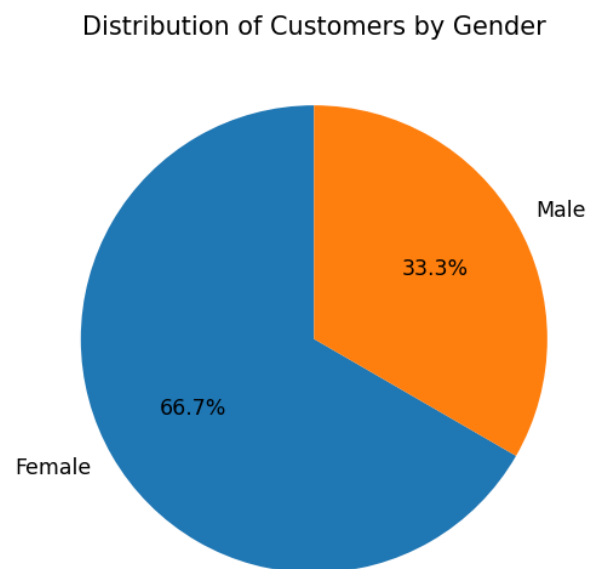
# -----
# 2. Pie Chart - Gender Distribution
# -----

gender_count = df["Gender"].value_counts()

plt.figure(figsize=(5,5))

plt.pie(gender_count,
        labels=gender_count.index,
        autopct="%1.1f%%",
        startangle=90)

```



```
plt.title("Distribution of Customers by Gender")
```

```
plt.savefig("Gender_Distribution_PieChart.png")
```

```
plt.show()
```

```
# -----
```

Customer Demographics Table

Customer ID	Age	Gender	Income (\$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

```
# 3. Table - Customer Demographics
```

```
# -----
```

```
fig, ax = plt.subplots(figsize=(7,2))
```

```
ax.axis('off')
```

```
table = ax.table(
```

```
    cellText=df.values,
```

```
    colLabels=df.columns,
```

```
    loc='center'
```

```
)
```

```
table.auto_set_font_size(False)
```

```
table.set_fontsize(10)
```

```
table.scale(1.2, 1.5)
```

```
plt.title("Customer Demographics Table")
```

```
plt.savefig("Customer_Demographics_Table.png")
```

```
plt.show()
```

Set 17: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.

```
# -----
```

```
# Set 17: Employee Performance Analysis
```

```
# -----
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# -----
```

```
# Create Dataset
```

```
# -----
```

```
data = {
    "Employee ID": [1, 2, 3],
    "Department": ["Sales", "HR", "Marketing"],
    "Years of Service": [5, 3, 7],
    "Performance Score": [85, 92, 78]
}
```

```
df = pd.DataFrame(data)
```

```
# Display Dataset
```

```
print("Employee Performance Dataset")
```

```
print(df)
```

```
# -----
```

```
# 1. Line Chart - Performance Trend
```

```
# -----
```

```
plt.figure(figsize=(6,4))
```

```
plt.plot(df["Years of Service"],
```

```
        df["Performance Score"],
```

```
        marker='o',
```

```
        linestyle='-',
```

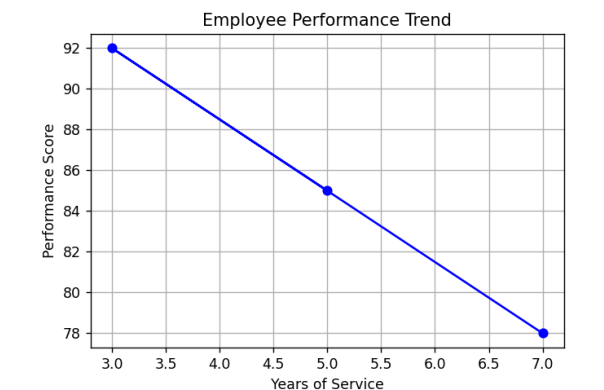
```
        color='blue')
```

```
plt.title("Employee Performance Trend")
```

```
plt.xlabel("Years of Service")
```

```
plt.ylabel("Performance Score")
```

```
plt.grid(True)
```



```
plt.savefig("Performance_Trend_LineChart.png")
```

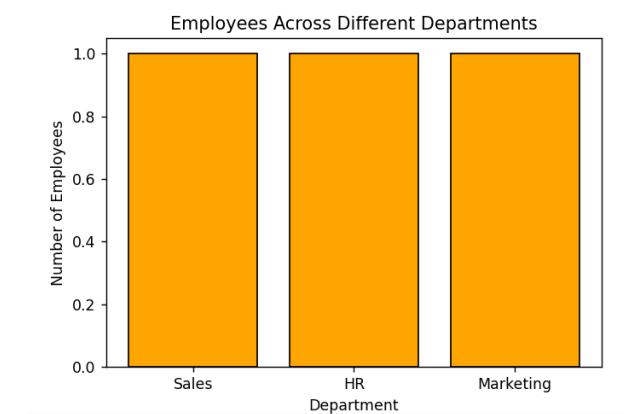
```
plt.show()
```

```
# -----
```

```
# 2. Bar Chart - Employees by Department
```

```
# -----
```

```
department_count = df["Department"].value_counts()
```



```
plt.figure(figsize=(6,4))
```

```
plt.bar(department_count.index,
```

```
        department_count.values,
```

```
        color="orange",
```

```
        edgecolor="black")
```

```
plt.title("Employees Across Different Departments")
```

```
plt.xlabel("Department")
plt.ylabel("Number of Employees")

plt.savefig("Department_BarChart.png")
plt.show()

# -----
# 3. Table - Employee Performance Data
# -----

fig, ax = plt.subplots(figsize=(8,2.5))
ax.axis('off')
```

```
table = ax.table(
    cellText=df.values,
    colLabels=df.columns,
    loc='center'
)
```

```
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1.2, 1.5)
```

```
plt.title("Employee Performance Table")
```

```
plt.savefig("Employee_Performance_Table.png")
plt.show()
```

Set 18: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available		Price (in \$)
1	Product A	250	20	
2	Product B	175	15	
3	Product C	300	18	

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a table to show the inventory data for each product. Label the table elements.

```
# -----
```

```
# Set 18: Product Inventory Management
```

```
# -----
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
# -----
```

```
# Create Dataset
```

```
# -----
```

```
data = {
```

```
    "Product ID": [1, 2, 3],
```

```
    "Product Name": ["Product A", "Product B", "Product C"],
```

```
    "Quantity Available": [250, 175, 300],
```

```
    "Price ($)": [20, 15, 18],
```

```
    "Category": ["Electronics", "Furniture",  
"Electronics"]
```

```
}
```

```
df = pd.DataFrame(data)
```

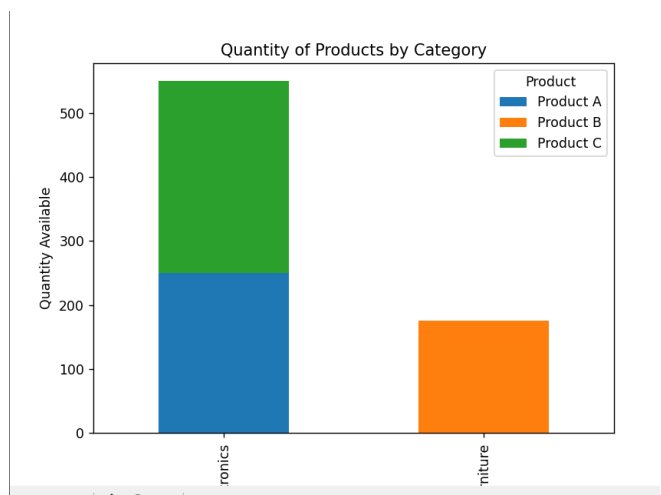
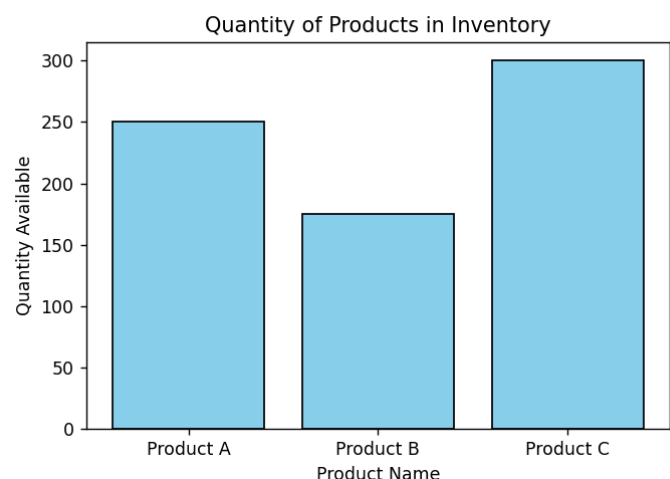
```
# Display Dataset
```

```
print("Product Inventory Dataset")
```

```
print(df)
```

```
# -----
```

```
# 1. Bar Chart - Product Quantity
```



```
# -----

plt.figure(figsize=(6,4))

plt.bar(df["Product Name"],
        df["Quantity Available"],
        color="skyblue",
        edgecolor="black")

plt.title("Quantity of Products in Inventory")
plt.xlabel("Product Name")
plt.ylabel("Quantity Available")

plt.savefig("Product_Quantity_BarChart.png")
plt.show()
```

```
# -----
# 2. Stacked Bar Chart - Quantity by Category
# -----
```

```
pivot = df.pivot_table(values="Quantity Available",
                        index="Category",
                        columns="Product Name",
                        fill_value=0)
```

```
pivot.plot(kind="bar",
           stacked=True,
           figsize=(7,5))
```

```
plt.title("Quantity of Products by Category")
plt.xlabel("Product Category")
plt.ylabel("Quantity Available")
plt.legend(title="Product")
```

```
plt.savefig("Stacked_BarChart.png")
plt.show()
```

```
# -----
```

```
# 3. Table - Inventory Data
```

```
# -----
```

```
fig, ax = plt.subplots(figsize=(8,2.5))
```

```
ax.axis('off')
```

```
table = ax.table(
    cellText=df.values,
    colLabels=df.columns,
    loc='center'
)
```

```
table.auto_set_font_size(False)
```

```
table.set_fontsize(10)
```

```
table.scale(1.2, 1.5)
```

```
plt.title("Product Inventory Table")
```

```
plt.savefig("Inventory_Table.png")
```

```
plt.show()
```

Set 19: Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.

```
# -----  
# Set 19: Survey Responses Analysis  
# -----
```

```
import pandas as pd  
import matplotlib.pyplot as plt
```

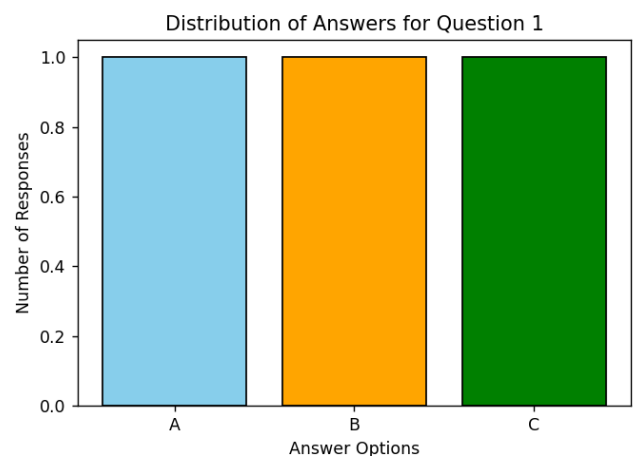
```
# -----  
# Create Dataset  
# -----  
data = {  
    "Survey ID": [1, 2, 3],  
    "Question 1": ["A", "B", "C"], "Question 2":  
    ["B", "A", "A"],  
    "Question 3": ["C", "D", "B"]  
}
```

```
df = pd.DataFrame(data)
```

```
# Display Dataset  
print("Survey Responses Dataset")  
print(df)
```

```
# -----  
# 1. Grouped Bar Chart - Question 1 Responses  
# -----
```

```
q1_counts = df["Question 1"].value_counts().sort_index()
```



```
plt.figure(figsize=(6,4))

plt.bar(q1_counts.index,
        q1_counts.values,
        color=["skyblue", "orange", "green"],
        edgecolor="black")

plt.title("Distribution of Answers for Question 1")
plt.xlabel("Answer Options")
plt.ylabel("Number of Responses")

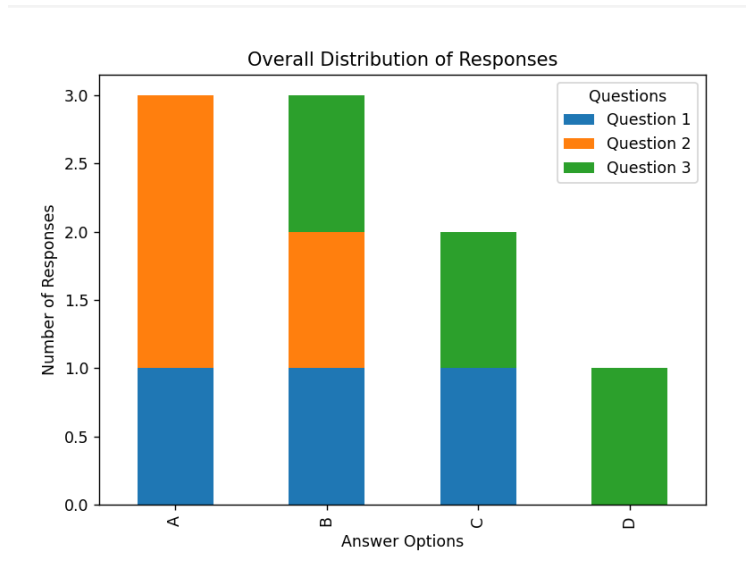
plt.savefig("Question1_GroupedBarChart.png")
plt.show()
```

```
# -----
# 2. Stacked Bar Chart - Overall Responses
# -----

response_counts = pd.DataFrame({
    "Question 1": df["Question
1"].value_counts(),
    "Question 2": df["Question
2"].value_counts(),
    "Question 3": df["Question
3"].value_counts()
}).fillna(0)
```

```
response_counts.plot(kind="bar",
                     stacked=True,
                     figsize=(7,5))

plt.title("Overall Distribution of Responses")
plt.xlabel("Answer Options")
plt.ylabel("Number of Responses")
```



```
plt.legend(title="Questions")

plt.savefig("Overall_Responses_StackedBarChart.png")
plt.show()
```

```
# -----
# 3. Table - Survey Responses
# -----

fig, ax = plt.subplots(figsize=(7,2.5))
ax.axis('off')
```

```
table = ax.table(
    cellText=df.values,
    colLabels=df.columns,
    loc='center'
)
```

```
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1.2, 1.5)
```

```
plt.title("Survey Responses Table")
```

```
plt.savefig("Survey_Responses_Table.png")
plt.show()
```

Set 20: Stock Analysis

Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118

Date	Stock A	Stock B	Stock C
2023-01-03	110	148	122

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.
2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.
3. Build a table to display the stock price data for each company over the given period. Label the table elements.

```
# -----
```

```
# Set 20: Stock Analysis
```

```
# -----
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# -----
```

```
# Create Dataset
```

```
# -----
```

```
data = {
    "Date": ["2023-01-01", "2023-01-02", "2023-01-03"],
    "Stock A": [100, 105, 110],
    "Stock B": [150, 152, 148],
    "Stock C": [120, 118, 122]
}
```

```
df = pd.DataFrame(data)
```

```
# Convert Date column to datetime format
```

```
df["Date"] = pd.to_datetime(df["Date"])
```

```
# Display Dataset
```

```
print("Stock Prices Dataset")
```

```
print(df)
```

```
# -----
```

```
# 1. Line Chart - Stock Prices
```

```
# -----
```

```
plt.figure(figsize=(8,5))
```

```
plt.plot(df["Date"], df["Stock A"], marker='o', label="Stock A")
```

```
plt.plot(df["Date"], df["Stock B"], marker='o', label="Stock B")
```

```
plt.plot(df["Date"], df["Stock C"], marker='o', label="Stock C")
```

```
plt.title("Stock Prices Over Time")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Stock Price ($)")
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.savefig("Stock_Prices_LineChart.png")
```

```
plt.show()
```

```
# -----
```

```
# 2. Bar Chart - Daily Percentage Change  
(Stock A)
```

```
# -----
```

```
# Calculate percentage change
```

```
df["Stock A % Change"] = df["Stock A"].pct_change() * 100
```

```
# Replace first NaN with 0
```

```
df["Stock A % Change"] = df["Stock A % Change"].fillna(0)
```

```
plt.figure(figsize=(7,4))
```



```
plt.bar(df["Date"].dt.strftime("%Y-%m-%d"),
        df["Stock A % Change"],
        color="orange",
        edgecolor="black")

plt.title("Daily Percentage Change in Stock A")
plt.xlabel("Date")
plt.ylabel("Percentage Change (%)")

plt.savefig("StockA_PercentageChange_BarChart.png")
plt.show()
```

```
# -----
```

```
# 3. Table - Stock Price Data
```

```
# -----
```

```
fig, ax = plt.subplots(figsize=(8,2.5))
ax.axis('off')
```

```
table = ax.table(
    cellText=df[["Date", "Stock A", "Stock B", "Stock
C"]].values,
    colLabels=["Date", "Stock A", "Stock B", "Stock C"],
    loc='center'
)
```

```
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1.2, 1.5)
```

```
plt.title("Stock Price Data Table")
```

```
plt.savefig("Stock_Prices_Table.png")
plt.show()
```

